

What Is a Computer System?

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A computer system consists of two parts: the software and the hardware. The software is the information in the form of data and program instructions. The hardware components are the electronic and mechanical parts of the system. The basic structure of a computer system is made up of three main hardware sections: the Central Processing Unit (CPU), the main memory, and the peripherals.

The CPU is a microprocessor chip which executes program instructions and coordinates the activities of all the other components. In a way, it is the "brain" of the computer. Larger computers may have two or more CPUs, in which case they are simply called "processors" because each is no longer a "central" unit. Nowadays, personal computers contain specialized graphic processors, with dedicated memory, for handling the computations needed to display complex graphics, such as for three-dimensional simulations and games.

The main memory holds the instructions and data which are currently being processed by the CPU. The internal memory of a microcomputer is usually composed of two sections: RAM (Random Access Memory) and ROM (Read Only Memory).

The peripherals are the physical units attached to the computer. They include input/output devices as well as storage devices. Input devices enable us to present information to the computer; for example, the keyboard and the mouse. Output devices allow us to extract the results from the computer; for example, we can see the output on the monitor or in printed form. Auxiliary data storage is usually provided by an internal hard disk and may be supplemented by other media such as flash drives or cloud storage services.

These are the main physical units of a computer system, generally known as the configuration.